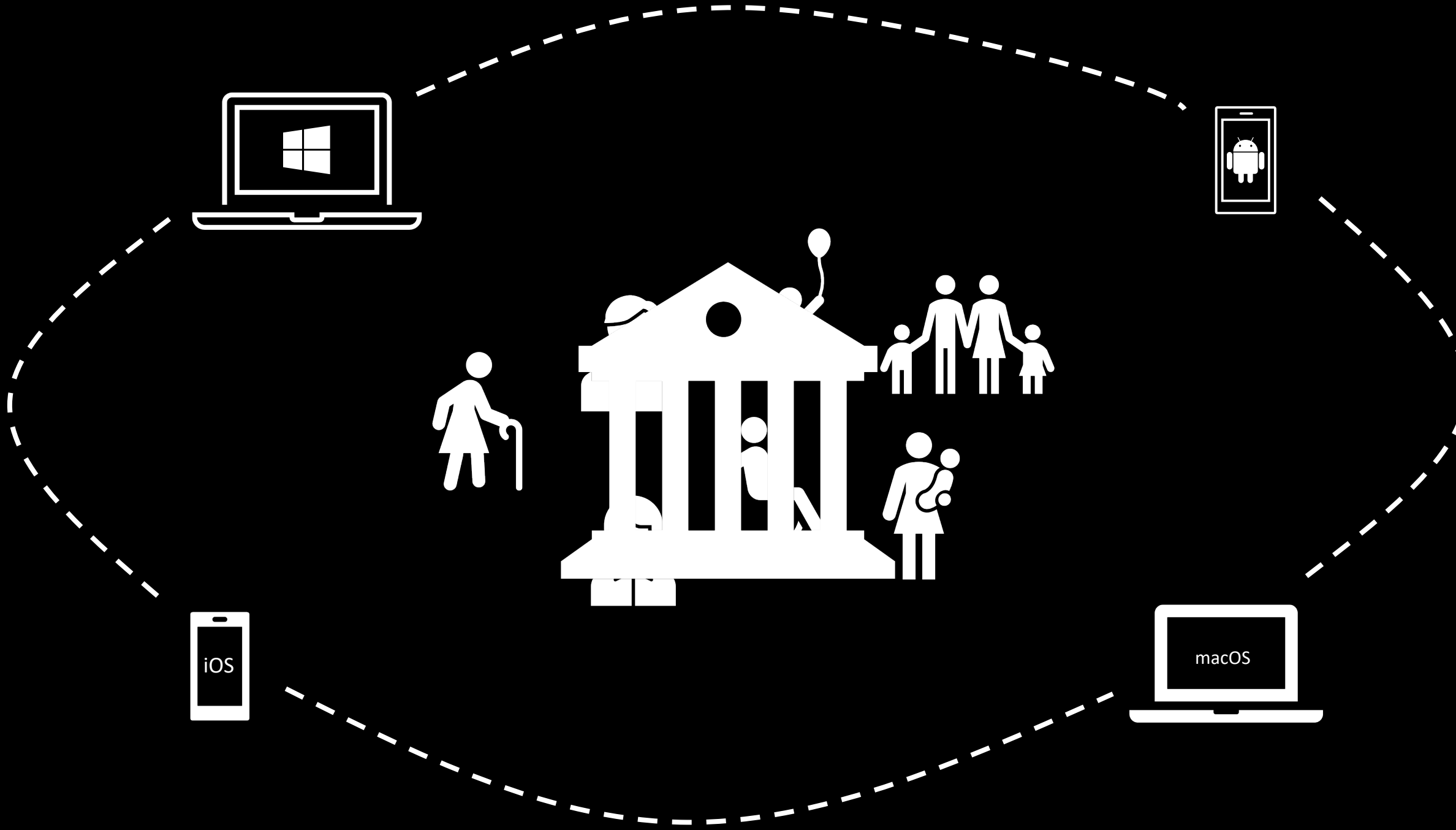
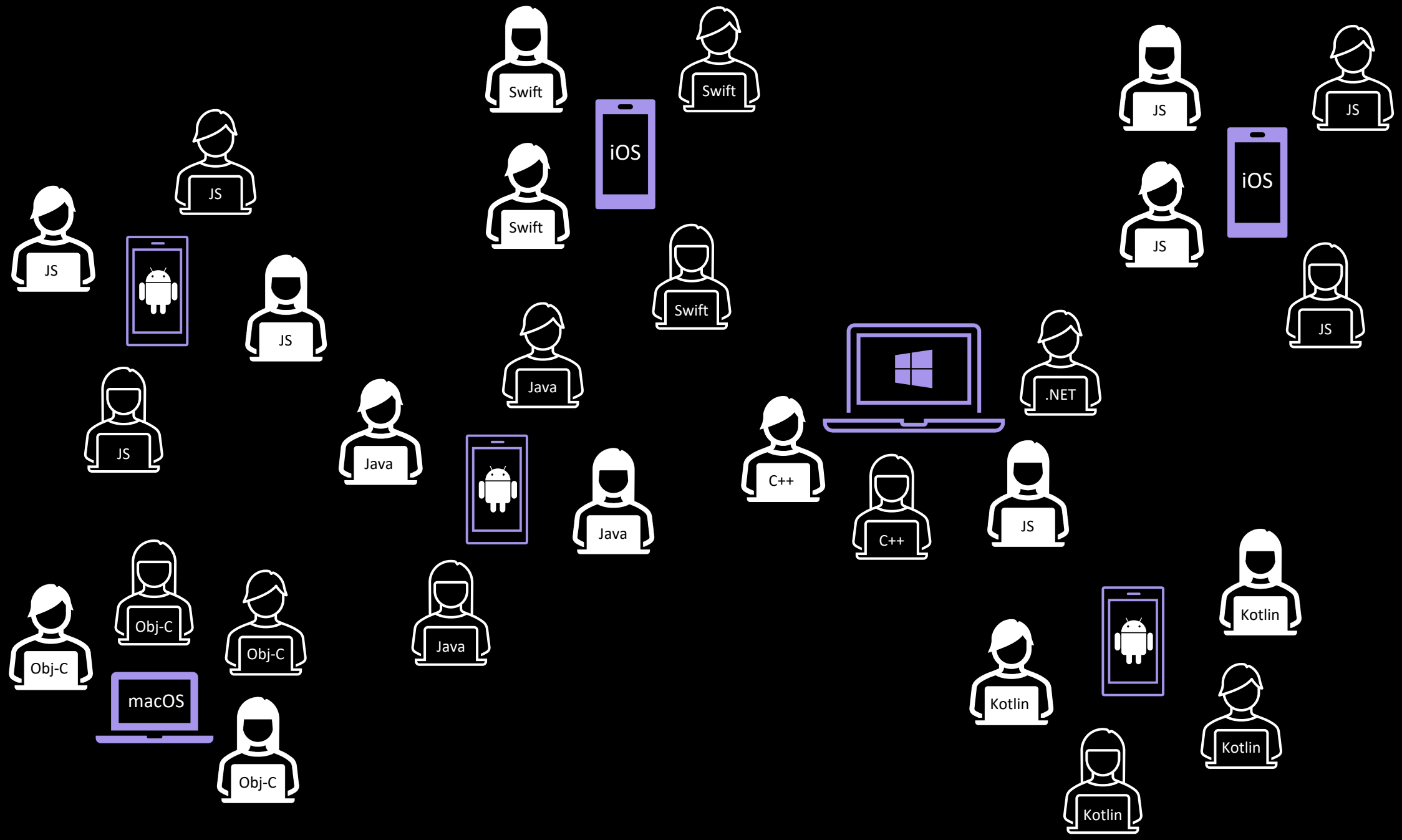
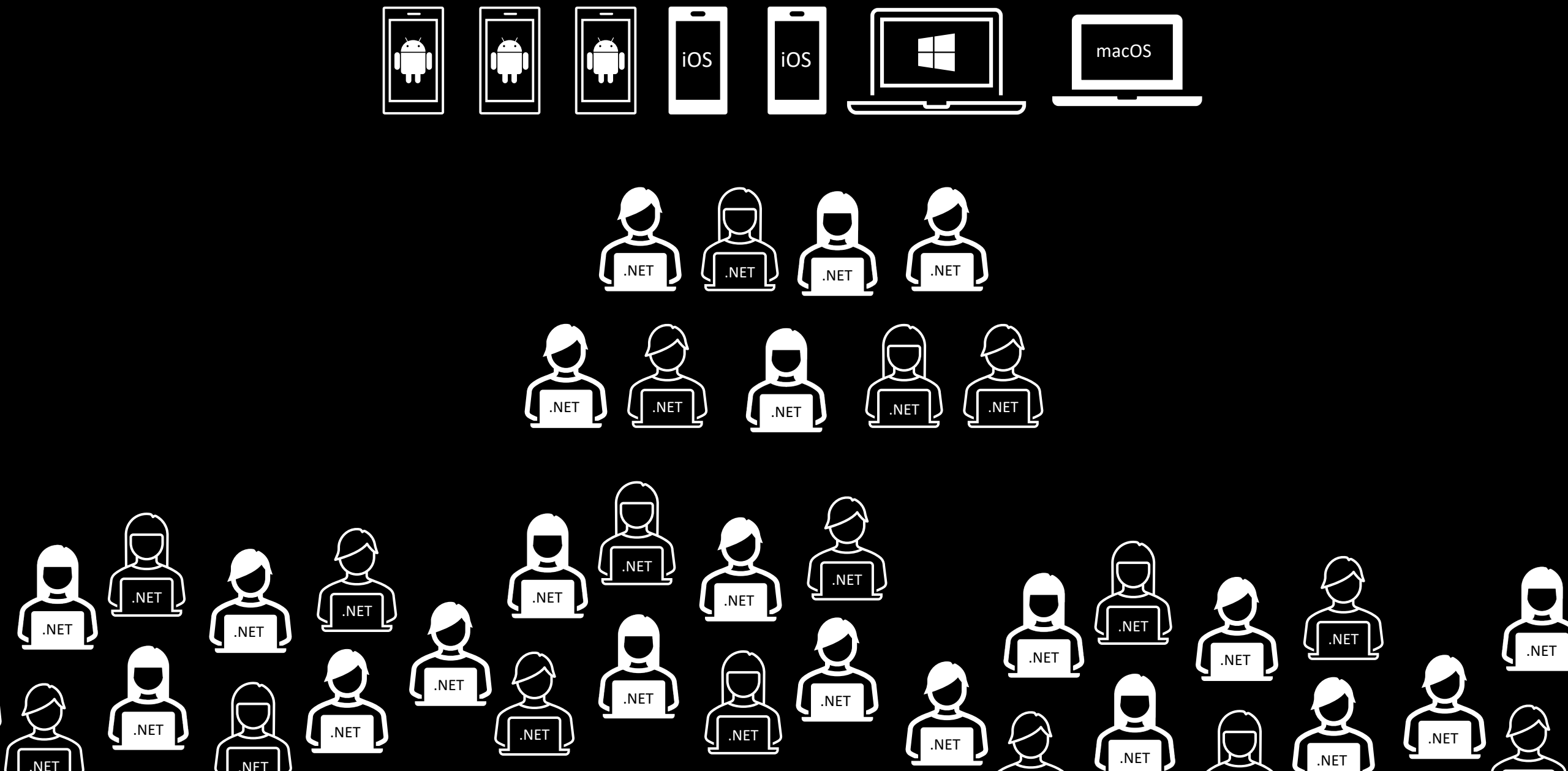
The Future of Native App Development in .NET 6

David Ortinau | Program Manager | @davidortinau
Maddy Leger | Program Manager | @maddyleger1

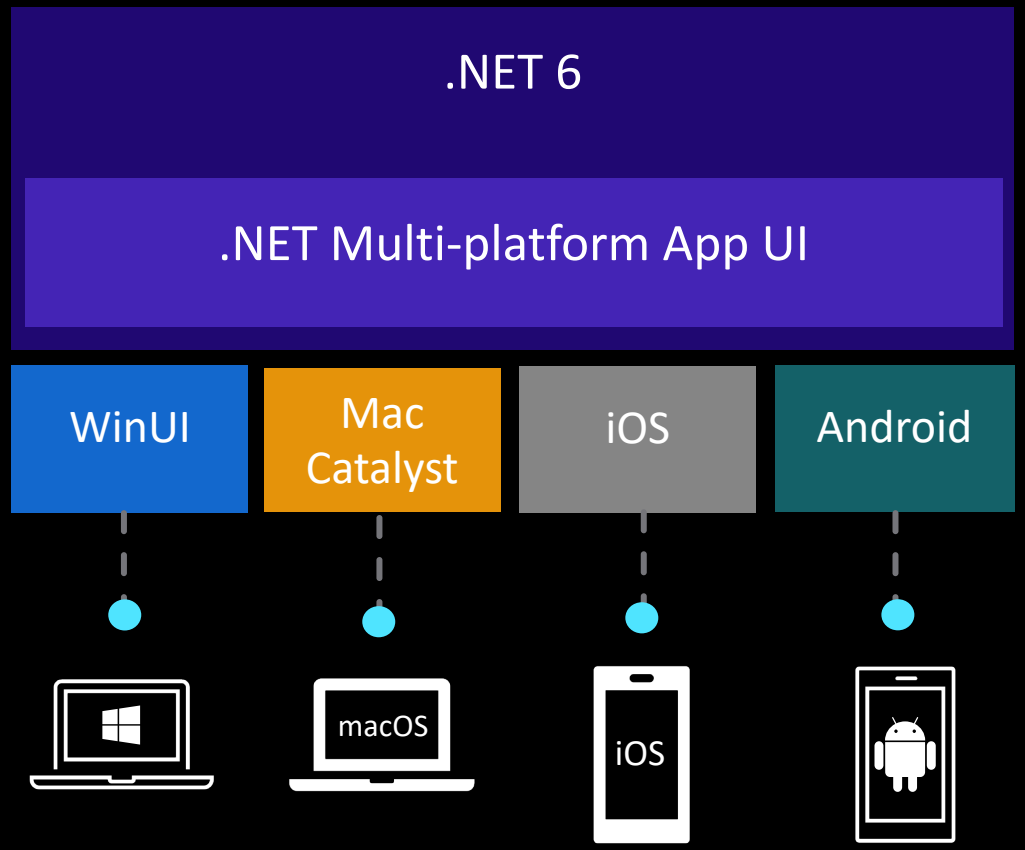



iOS
macOS


Swift
Swift
iOS
Swift
Swift
JS
JS
JS
JS
JS
Java
Java
Java
Java
C++
C++
.NET
JS
JS
iOS
JS
JS
Obj-C
Obj-C
Obj-C
Obj-C
macOS
Kotlin
Kotlin
Kotlin
Kotlin


iOS
iOS
macOS
.NET

.NET Multi-platform App UI



Cross-platform, native UI

Single project system, single codebase

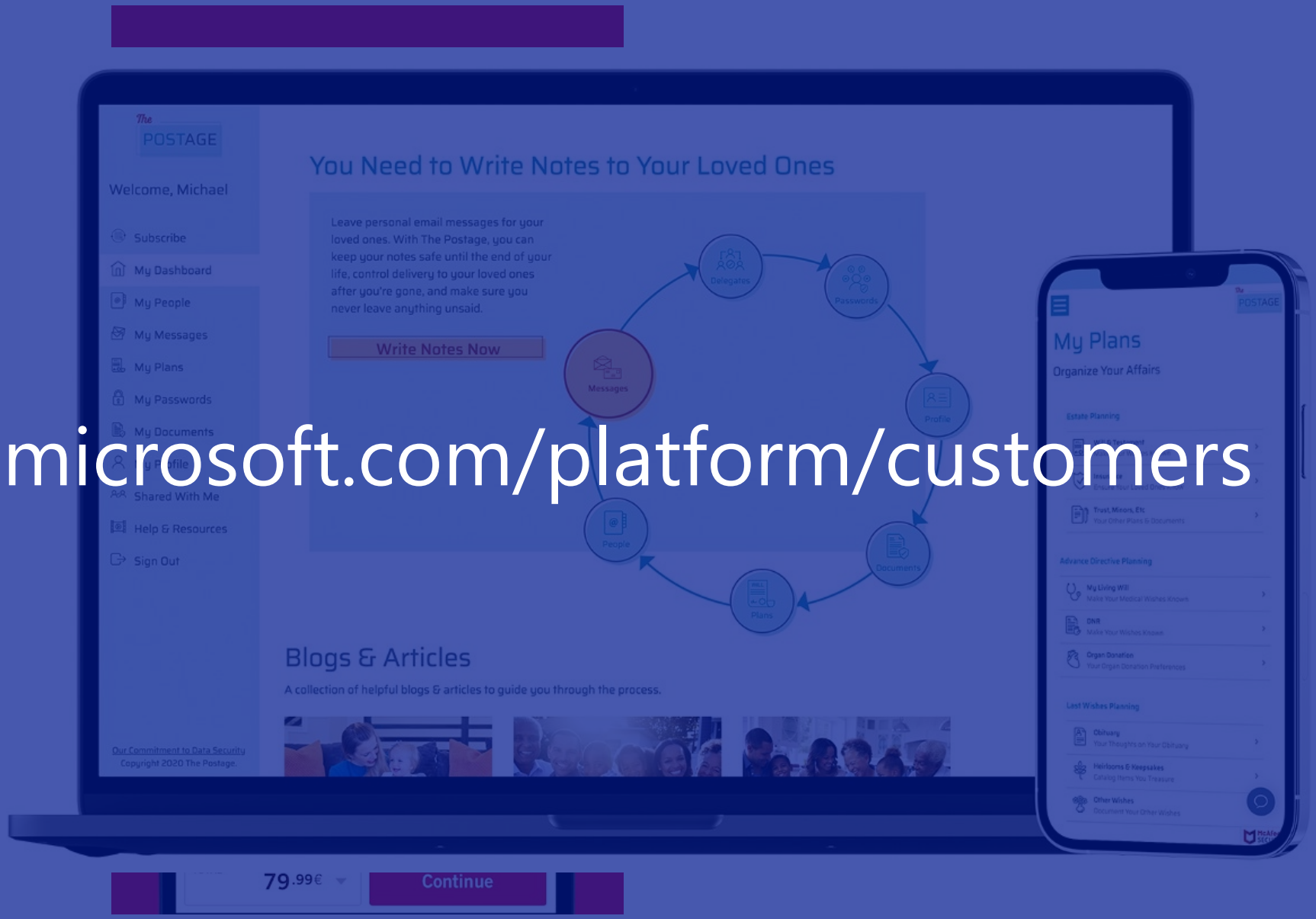
Deploy to multiple devices, mobile & desktop

General availability in .NET 6

github.com/dotnet/maui

.NET Multi-platform App UI is the evolution of Xamarin.Forms extended from mobile to desktop scenarios with UI controls rebuilt from the ground up for performance and extensibility.

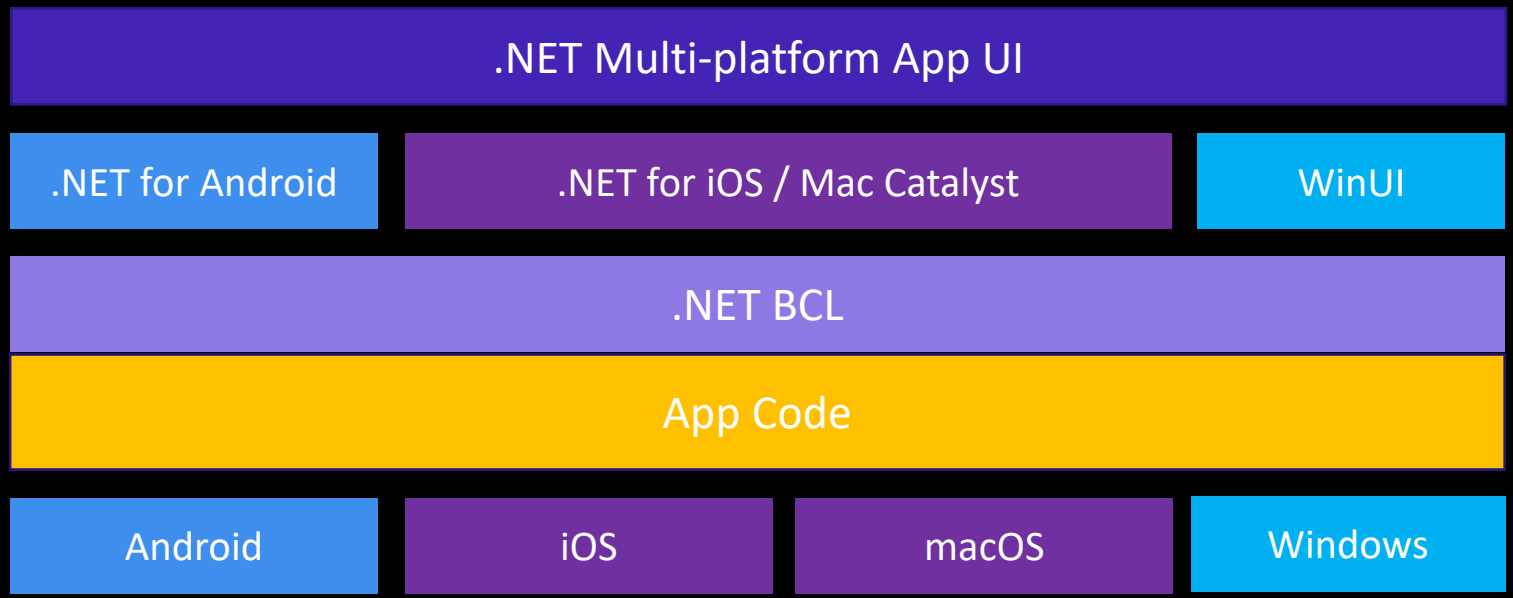
https://dotnet.microsoft.com/platform/customers



Pillars of Client App Development



How is this native?



Pages



ContentPage MasterDetailPage NavigationPage TabbedPage TemplatedPage

Layouts



StackLayout AbsoluteLayout RelativeLayout Grid FlexLayout

Controls

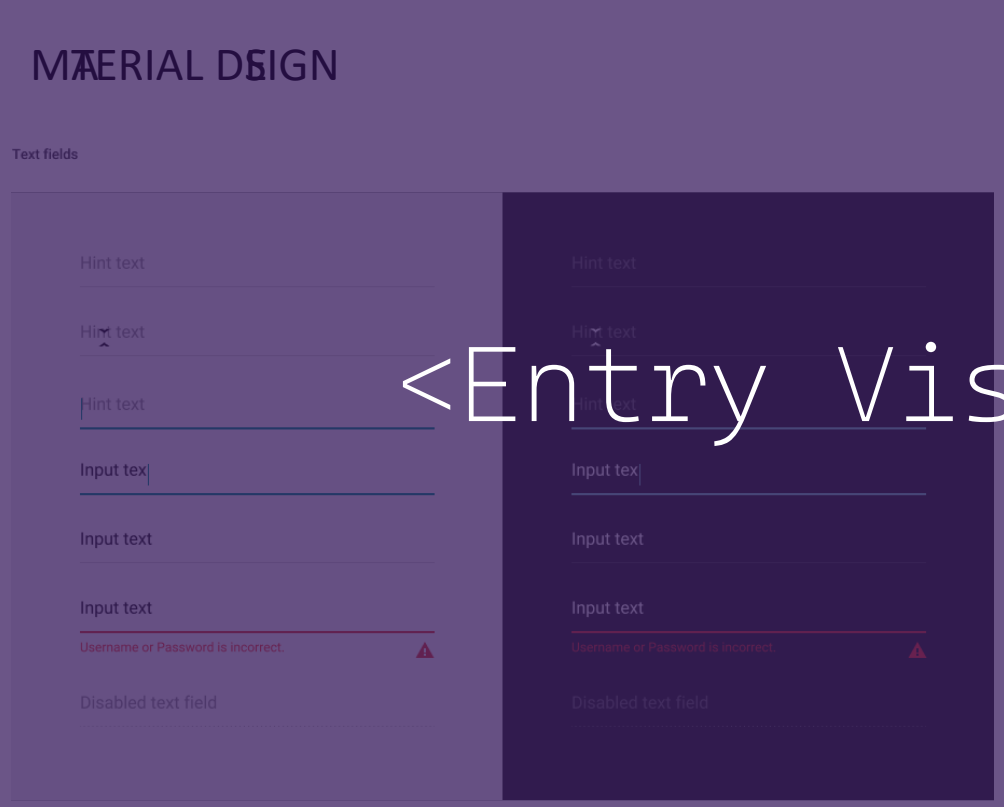
CarouselView
CollectionView
RefreshView
IndicatorView
Checkbox
RadioButton
ImageButton
SwipeView
ActivityIndicator
BoxView
Button
DatePicker
Editor
Entry
Image
Label
ListView
Map
OpenGLView
Picker
ProgressBar
SearchBar
Slider
Stepper
TableView
TimePicker
WebView
EntryCell
ImageCell
SwitchCell
TextCell
ViewCell
Path
Ellipse
Polygon
Rectangle
Line
Polyline

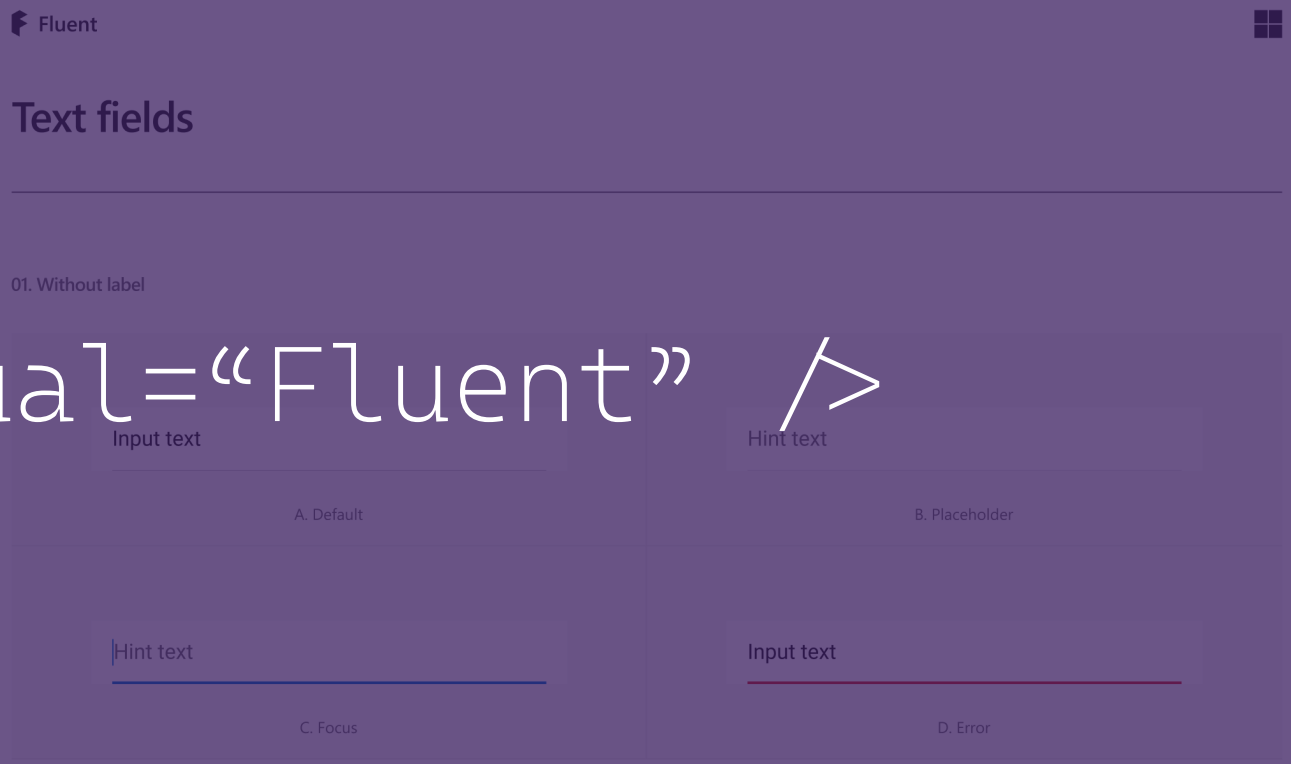
Native UI

Elements are realized via platform handlers.

	ContentPage	Label	Picker
Android	Activity / Fragment	TextView	AlertDialog + LinearLayout + EditText + NumberPicker
iOS/Mac	UIViewController	UILabel	UITextField + UIPickerView + UIToolbar + UIBarButtonItem
WinUI	Page	TextBlock	ComboBox

Visual Design





<Entry Visual="Fluent" />

Platform Features

AppThemes (Dark Mode)

Accessibility

Adaptive Triggers

Brushes – Solid & Gradient

Clipping

Cross-platform Lifecycle

CSS & Styles

Databinding (Compiled, Multi, Relative)

Dual Screen Support

Essentials (Sensors, Permissions, Sharing, etc.)

FontImageSource

Gestures

MessagingCenter

Microsoft.Extensions.*

Multi-Window

Navigation – URI based

Right-to-Left

Visual State Manager

XAML

Component Vendors

















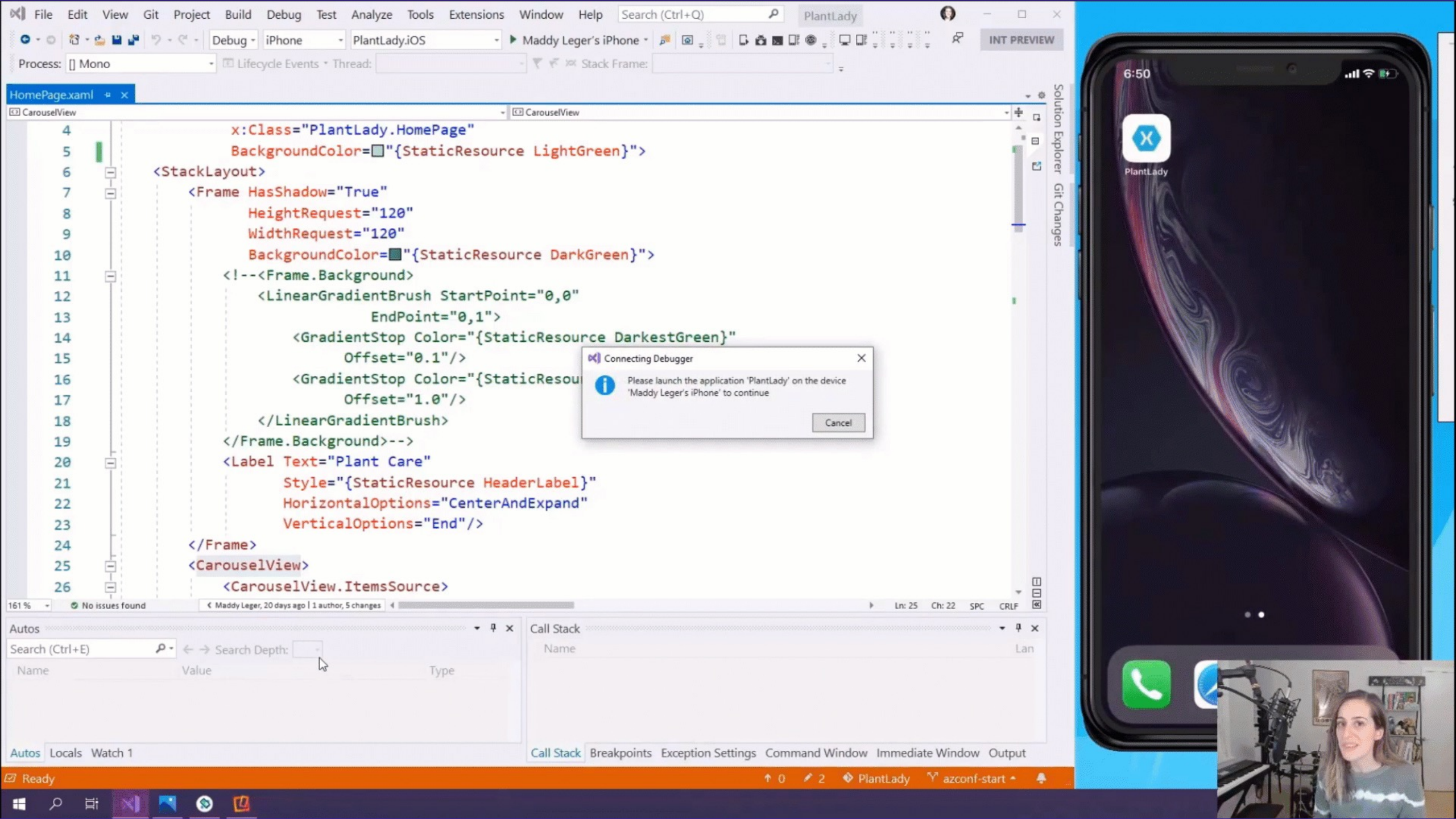


Developer Experience

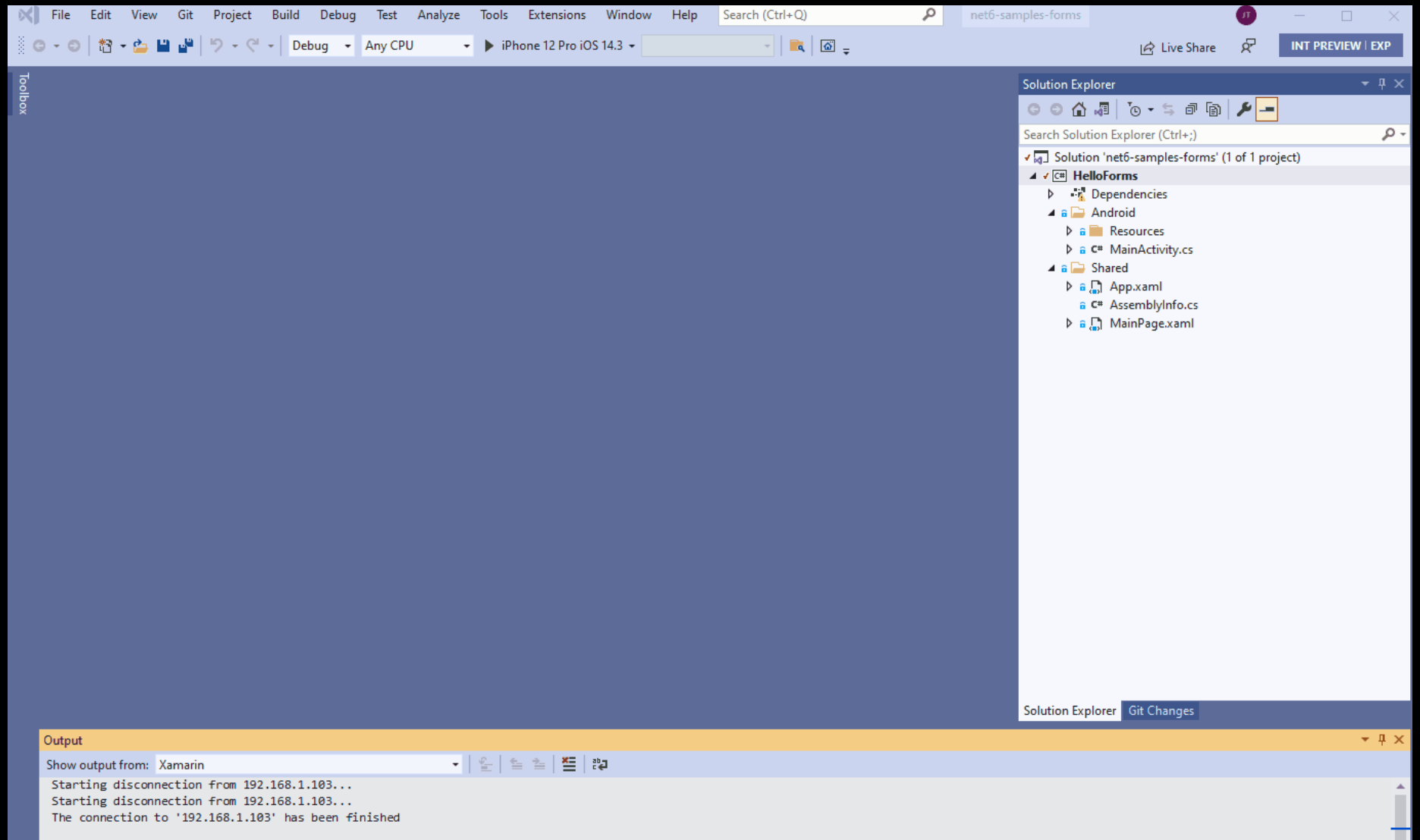


Demo

Familiar tools, familiar environment


File Edit View Git Project Build Debug Test Analyze Tools Extensions Window Help
Search (Ctrl+Q)
PlantLady
Debug iPhone PlantLady.iOS Maddy Leger's iPhone
INT PREVIEW
Process: [] Mono
Lifecycle Events Thread:
Stack Frame:
HomePage.xaml
CarouselView
CarouselView
4 x:Class="PlantLady.HomePage"
5 BackgroundColor="{StaticResource LightGreen}">
6 <StackLayout>
7 <Frame HasShadow="True"
8 HeightRequest="120"
9 WidthRequest="120"
10 BackgroundColor="{StaticResource DarkGreen}">
11 <!--<Frame.Background>
12 <LinearGradientBrush StartPoint="0,0"
13 EndPoint="0,1">
14 <GradientStop Color="{StaticResource DarkestGreen}"
15 Offset="0.1"/>
16 <GradientStop Color="{StaticResou
17 Offset="1.0"/>
18 </LinearGradientBrush>
19 </Frame.Background>-->
20 <Label Text="Plant Care"
21 Style="{StaticResource HeaderLabel}"
22 HorizontalOptions="CenterAndExpand"
23 VerticalOptions="End"/>
24 </Frame>
25 <CarouselView>
26 <CarouselView.ItemsSource>
Connecting Debugger
Please launch the application 'PlantLady' on the device 'Maddy Leger's iPhone' to continue
Cancel
161 %
No issues found
Maddy Leger, 20 days ago | 1 author, 5 changes
Ln: 25 Ch: 22 SPC CRLF
Autos
Search (Ctrl+E)
Search Depth:
Name Value Type
Call Stack
Name Lan
Autos Locals Watch 1
Call Stack Breakpoints Exception Settings Command Window Immediate Window Output
Ready
0 2 PlantLady azconf-start
Solution Explorer Git Changes
6:50
PlantLady

Simplified Cross-Platform Development

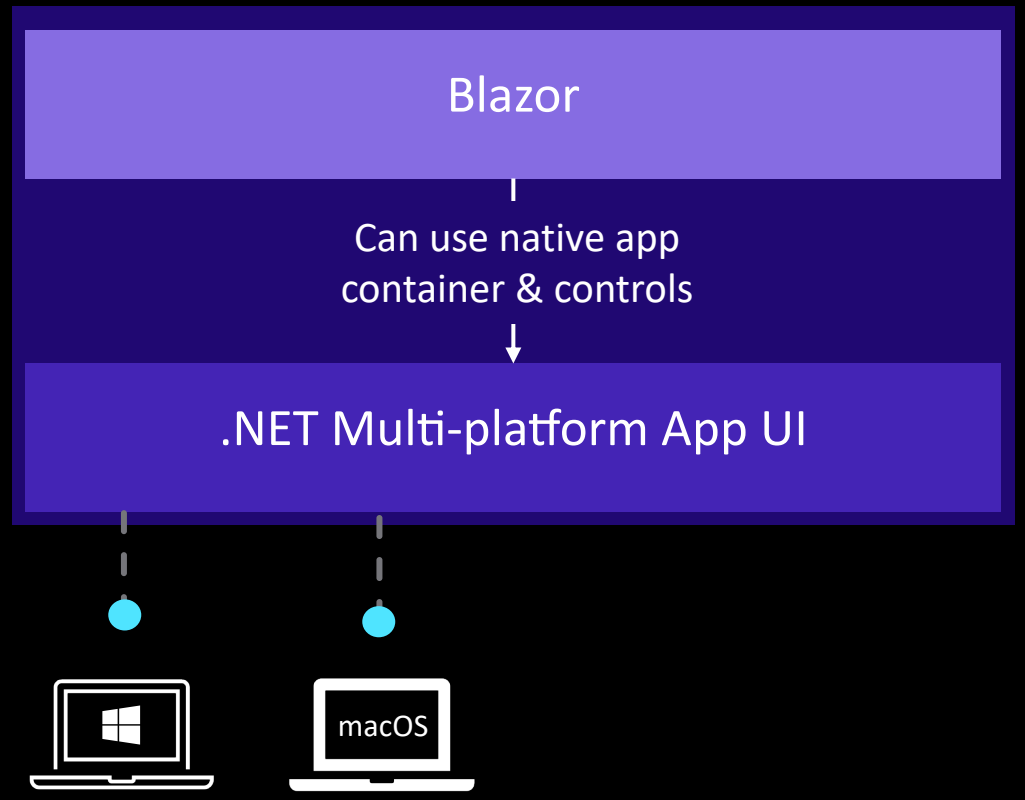


Single place for:

- Images/Assets
- App icons
- Fonts
- Launch screens
- Manifest
- Startup/App.cs

...and more!

Blazor hybrid desktop apps

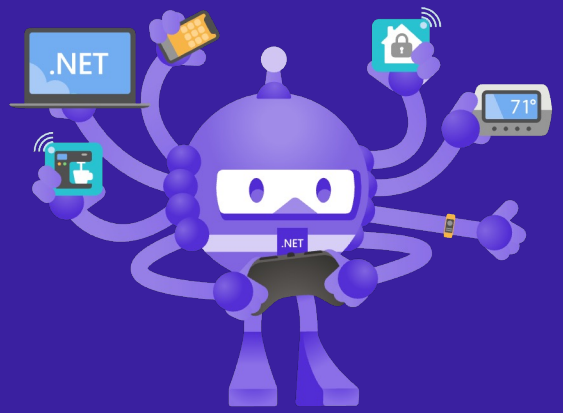


Reuse UI components across native and web

Built on top of .NET Multi-platform App UI

Native app container & embedded controls

General availability in .NET 6



Get Ready!

Timelines and how you can contribute

.NET 6 Timeline

Preview 1

- Includes
  - Runtime and BCL, Mobile Workloads
- Audience
  - Contributors

Preview 2

- Includes
  - .NET MAUI views, layouts, everything that’s been merged
  - Android, iOS, macOS
- Audience
  - Contributors

.NET 6 Timeline

Preview 3 & 4

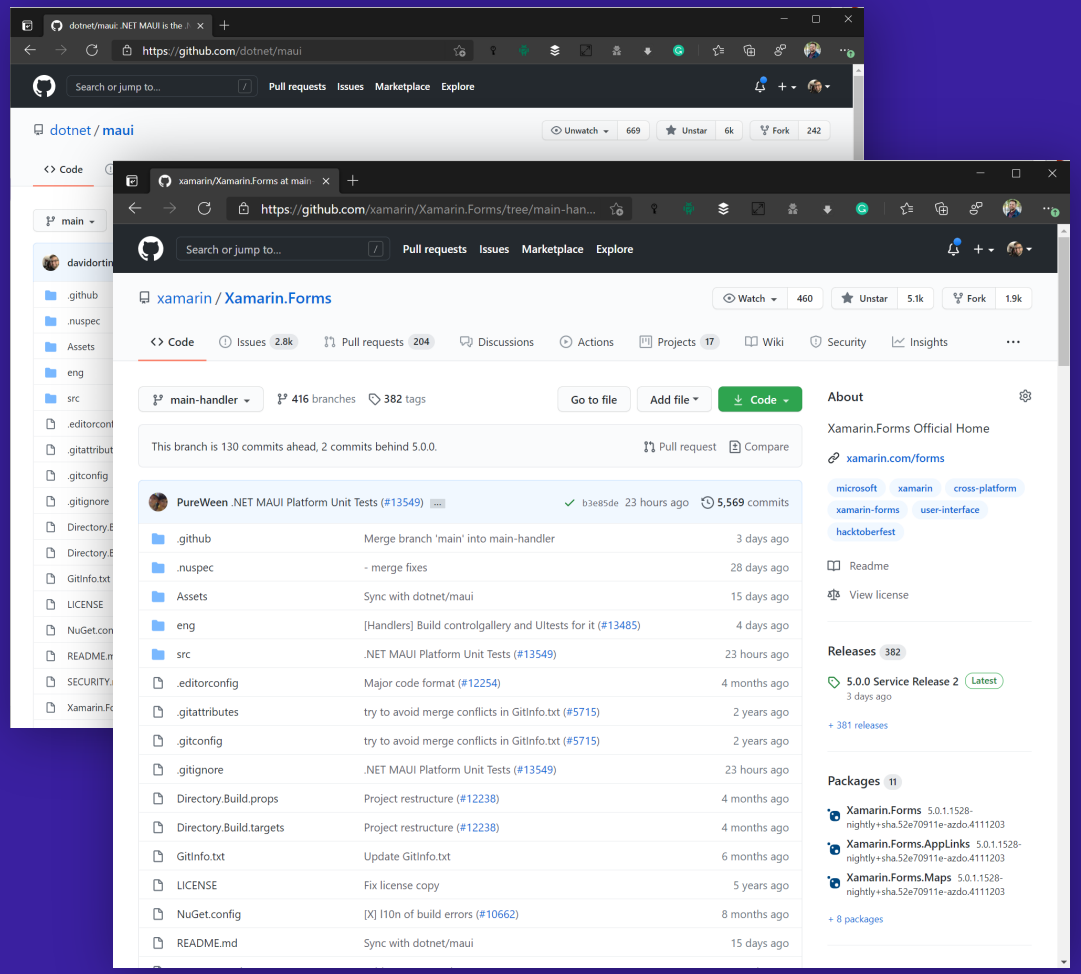
- Includes
  - .NET MAUI will have most controls, layouts, pages, and services
  - Adding WinUI 3
- Audience
  - You!
- What should you do with this release?
  - Build new apps and kick the tires
  - Open issues and join discussion on Discord

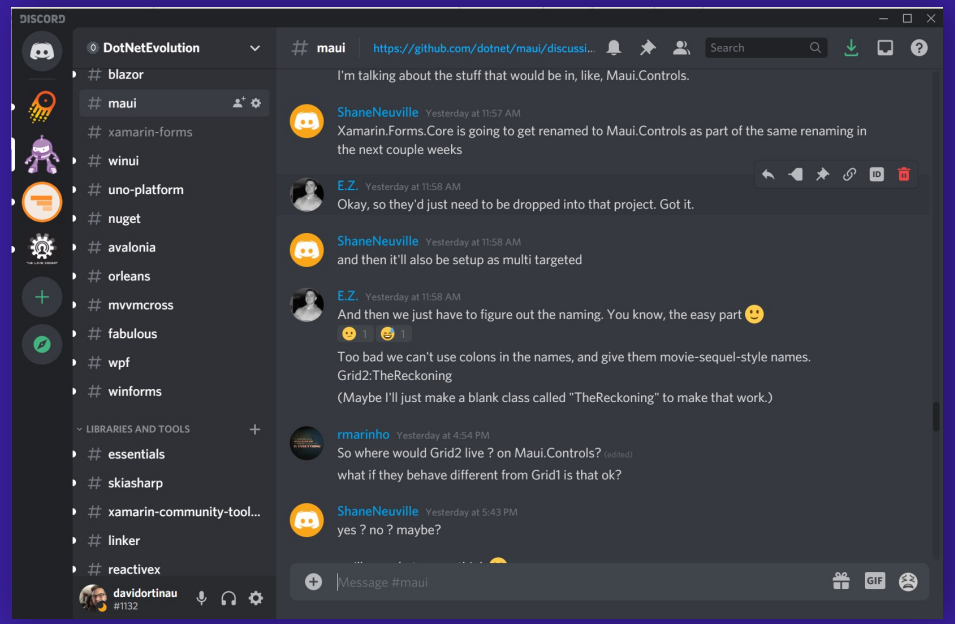
.NET 6 Timeline

Preview 5 – Release Candidate

- Includes
  - .NET MAUI fills the remaining gaps, responds to feedback
- Audience
  - Everyone!
- What should you do with this release?
  - Build new apps and kick the tires
  - Migrate existing apps
  - Open issues and join discussion on Discord

Where is this happening?

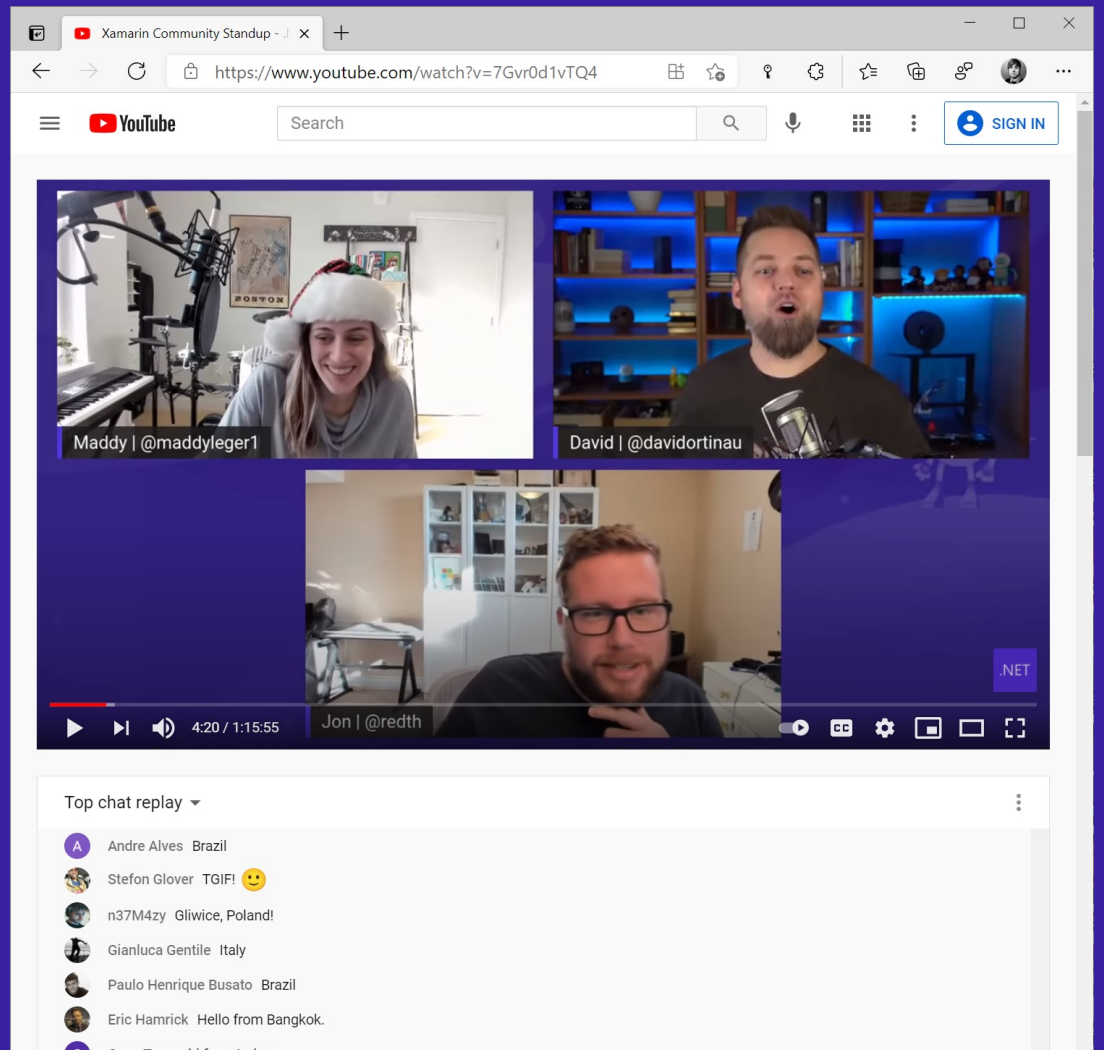




https://aka.ms/discord-maui

https://github.com/dotnet/maui

.NET Community Standup



https://live.dot.net

1st Thursday of the Month @ Noon PT / 3 ET

- Community News
- Contributions
- .NET MAUI Status Report
- Shenanigans!

OSS Library Maintainer - Monthly Sync

- 1st Tuesday of the Month @ Noon PT / 3 ET
- Teams Call to deep dive on architecture and ask ?s
- Current representation from:
  - FreshMvvm
  - Mobile Build Tools
  - MvvmCross
  - Prism
  - ReactiveUI
  - Shiny
  - Xamarin Community Toolkit
  - Uno Platform

To join:
david.ortinau@microsoft.com

Thanks for joining!

David Ortinau | Program Manager | @davidortinau

Maddy Leger | Program Manager | @maddyleger1

